

wireless technology, more and more parameters become vital. If the user is working on TD-LTE, the main parameters become demodulation features and parameters like RSRP, RSRQ and SINR.

Baldua: RF/wireless signals are very common in the present communication media.

Mobile phones, wireless data cards, satellite signals in the form of DTH, wireless communication sets in defence and police wireless systems are known to all of us.

Fundamentally, signal levels (amplitude and frequency of RF communication), power of transmitting channels, adjacent channel power, and channel noise are regular and very common measurements in RF.

Narasimhan: The most standard and common parameters that need to be tested for RF/wireless applications include spectrum power (power spectrum, spectral density, power spectral density or energy spectral density) along with channel power, quality of modulation (modulation index, etc), signal-to-noise ratio and emissions (including broadcasting, low-speed and two-way emissions).

Kumar: For any RF or wireless device, transmitter and receiver measurements are a standard. Transmit and receive parameters can reduce the performance of wireless systems, or even prevent RF devices from working together. Typical measurements in this domain are transmit power, EVM, power spectral density, spectral emission mask, sensitivity and packet error rate.

Specifically in wireless testing, it is important to test coexistence of WLAN/BT as these share the same ISM band.

Could you list some of the latest trends in the field of WLAN testing?

Samant: As 802.11ac penetrates the market, we will see more and more 'plugged-in' devices using advanced MIMO techniques such as beam-forming and multi-user MIMO. The use of beam-forming was limited in 802.11n standard because of inclusion of both implicit and explicit beam-forming definitions in the standard. Along with that, low-density parity checking (LDPC) codes improve the coding gain by nearly 2 dB. This would improve performance in noisy situations or increase the range. Most vendors are expected to support LDPC in 802.11ac.

Tripathi: Some of the trends we have noticed include the combination of several equipment in the same box and on a single platform. A common example would be spectrum analyser, vector network analyser and power meter all available in the same box from a vendor.

Benchtop instruments are also making their appearance in a big way, replacing the traditional table-top variants in most applications.

Baldua: Today, we see a lot of test instruments with automatic test capability, which are also capable of displaying the results in a variety of different units to make comprehensibility a whole easier. Also, the whole test and measurement environment is now starting to adjust to the

presence of 802.11ac standard and trying to make products that can serve the new-generation chips so that it is not left behind the competition.

Narasimhan: Today, RF needs to be understood as a dynamic phenomenon. Current communication systems operate in burst mode, which requires understanding of this burst behaviour. There is also a lot of spectral reuse, which can lead to interference. In satellite domain there is a lot of malicious transmission hidden within a signal to disrupt the signal quality. Using a conventional test method, the system might pass all tests and work perfectly in the lab. But to work in the market, a device needs robust testing with the aid of real-time spectrum analysers.

Kumar: Recently, the demands for even higher data rates on WLAN technologies have expanded applications to wireless HD video streaming among TV sets, multimedia players, video cameras, PCs, smartphones and tablets. IEEE 802.11ac aims for a minimum 500Mbps data rate and up to around 6.93 Gbps with various advanced technologies including wider channel bandwidth up to 160 MHz, and higher modulation up to 256 QAM.

How is the industry moving towards increasing the throughput of WLAN?

Samant: One way is higher modulation schemes like 256 QAM and MIMO. So the trend that I am seeing is a cross-layer collaboration between mac and physical layers so that more information is shared between these two rather than increasing the complexity of the chip.

Tripathi: The industry is moving towards achieving higher data rates for WLAN devices. End users want faster delivery of content and a higher data rate is the most demanding expectation.

Now that 802.11 a, b, g and n standards have matured, work is underway for 802.11ac and 802.11ad, which are targeted to provide 7000Mbps per stream data rates.

Baldua: The throughput of WLAN completely depends on the configuration and the design associated with it. It is a built-in system with its own properties and tendencies. As time passes, newer configurations will enable designers to radically improve the existing throughput values and move towards a higher level for them.

Narasimhan: With 802.11ac already being implemented, MIMO is now the latest way to increase your throughput. The industry is now moving towards 4x4 and 8x8 MIMO to adjust to the new standards. Once you have a 2x2 MIMO system, you are basically pumping twice the data in the same bandwidth, and this doubles your throughput.

Kumar: The latest WLAN technology, 802.11ac, builds on 802.11n with a wider RF bandwidth (up to 160 MHz), MIMO and high-density 256QAM modulation promises to take the throughput of WLAN to newer places. As a result of the advanced technology, WLAN 802.11ac will provide a high throughput of 1 Gbps below 6 GHz, for multiple stations. ●

(The complete interview can be read at electronicsforu.com)